

From Answer Voting to Claim Auditing: Objective-Aware Hallucination Control for RAG under Corrupted Evidence

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Abstract

Retrieval-augmented generation (RAG) systems use retrieved evidence to ground model outputs and reduce hallucination, yet a plausible or even correct final answer may still preserve unsupported, contradicted, or fabricated local claims in its evidence trace. This creates a claim-level robustness problem, especially in multi-agent RAG settings where some evidence may be corrupted. Prior work on multi-agent debate and social-deduction-style hallucination detection addresses this problem through answer selection or unreliable agent detection, but neither directly determine which individual claims should survive. Is hallucination-resistant RAG best treated as answer selection, agent filtering, or claim-level auditing? Our starting point is open-rubric evaluation: if accuracy-only scoring rewards confident guesses, then a safer RAG system should make the costs of wrong answers, unsupported claims, unnecessary abstention, deleted evidence, and audit effort explicit. We study this question through OR-Wolf, an objective-aware audited claim game for multi-agent RAG. Agents submit atomic claims with evidence quotes and confidence scores, other agents challenge concrete claim IDs, and a verifier settles claim-evidence support. An explicit designer objective then controls claim retention by balancing bad-claim survivals, wrong non-abstained answers, false abstentions, deleted good evidence, and audit cost. Experiments on 200 SciFact examples show that answer-level methods can leave contaminated claims in the retained evidence trace, while MUG-style agent filtering provides a strong but coarse baseline. In contrast, objective-aware OR-Wolf variants make the safety-coverage trade-off explicit by filtering unsupported claims while preserving useful supported claims when evidence coverage remains sufficient. Our findings therefore provide a more precise perspective on hallucination mitigation in RAG: unreliable content is not best addressed only by voting over answers or removing speakers, but by auditing which claims are allowed to survive.

Introduction

Hallucination remains a central challenge for LLMs in evidence-grounded reasoning, where unsupported statements can undermine the reliability of an otherwise plausible response (Ji et al. 2023). Retrieval-augmented genera-

tion (RAG) has become a widely adopted approach to this problem by using externally retrieved evidence to constrain model outputs (Lewis et al. 2020). However, retrieval alone does not guarantee factual generation. A generated response typically consists of multiple claims, where each claim is an individual factual statement that can be assessed against the retrieved evidence. Even when relevant evidence is provided, some claims may still be unsupported, contradicted, or fabricated (Niu et al. 2024). If such claims are preserved, they can contaminate the reasoning process, mislead downstream verification, and reduce the factual reliability of the final response.

This claim-level view exposes a gap in how evidence-grounded generation is evaluated. Prior work on fine-grained factuality shows that generated text often mixes supported and unsupported atomic facts, making coarse binary judgments inadequate (Min et al. 2023). Work on attribution and citation-based generation further separates answer correctness from source support: a response may be correct overall while some of its statements lack support from the cited evidence (Rashkin et al. 2023; Gao et al. 2023). Similar concerns appear in RAG evaluation: RAGAS and ARES evaluate faithfulness or answer faithfulness as dimensions distinct from answer relevance or correctness (Es et al. 2024; Saad-Falcon et al. 2024), while RAGTruth shows that RAG outputs can still contain unsupported or contradictory content even when retrieved evidence is available (Niu et al. 2024). Together, these findings suggest that final-answer accuracy can hide claim-level contamination.

This evaluation gap is not only diagnostic; it also affects how systems are optimized. Recent work in Nature argues that accuracy-based evaluation can incentivize hallucination by rewarding guessing over admitting uncertainty (Kalai et al. 2026). Related theoretical work further shows that language models can produce plausible false facts even under calibration, especially for facts that are difficult to infer reliably from training data (Kalai and Vempala 2024). In RAG, this incentive problem can appear below the answer level: when a model is rewarded for producing a confident answer but is not sufficiently penalized for unsupported intermediate statements, it may fill evidential gaps with plausible local claims rather than abstaining or marking uncertainty. This suggests that hallucination control should not optimize answer accuracy alone, but should also account for unsupported claim

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survival, false abstention, deletion of supported evidence, and auditing cost.

Multi-agent methods provide a natural starting point because evidence corruption can affect different agents differently. By evidence corruption, we mean that retrieved documents may contain misleading, contradictory, incomplete, or counterfactually modified information, causing some agents to generate claims that appear plausible from their local context but are not globally supported. Multi-agent debate improves reliability by letting multiple model instances propose and discuss answers before aggregation (Du et al. 2024). A closer neighbor is Multi-Agent Undercover Gaming (MUG), which introduces counterfactual tests and social-deduction-style interaction to identify hallucinating agents (Liang, Wei, and Zheng 2026). These methods matter because they move multi-agent reasoning from passive agreement toward active verification: disagreements induced by asymmetric or corrupted evidence become observable and testable.

However, answer-level debate and agent-level filtering remain too coarse for hallucination-sensitive RAG. They ask which answer should win, or which agent should be trusted, rather than which individual claims should be retained. This distinction matters because supported and unsupported claims can coexist in the same response. A generally unreliable agent may still contribute a useful evidence-grounded claim, while a generally reliable agent may still introduce an unsupported local claim. Therefore, removing an entire agent can discard good evidence, while keeping the agent can preserve bad claims. The unit of adjudication should be the claim, not the speaker.

To address this issue, we propose OR-Wolf, an objective-aware claim auditing framework for hallucination-resistant RAG under corrupted evidence. OR-Wolf borrows the social-deduction intuition of hidden unreliable information and public challenge, but uses it for claim auditing rather than role identification. Agents decompose their responses into atomic claims, attach evidence quotes and confidence scores, and record these claims in a shared ledger. Other agents may challenge suspicious claim IDs, and a verifier settles each claim as supported, contradicted, unsupported, fabricated, or insufficiently grounded with respect to the retrieved evidence. Objective-aware retention then decides which claims survive, and the final answer is generated only from retained claims. This shifts hallucination control from answer voting to claim survival control.

OR-Wolf also makes the safety–coverage trade-off explicit. A system optimized only for final-answer accuracy may keep false claims when the final answer is still correct. Conversely, a system that filters too aggressively may avoid hallucination by abstaining too often or by deleting useful evidence. OR-Wolf therefore defines a designer objective that penalizes wrong non-abstained answers, retained bad claims, false abstentions, deleted good evidence, and audit cost. This objective specifies the system-level loss to be minimized, while agent payoffs provide the incentive interface for claim submission, challenge, and calibrated confidence reporting. Unsupported claims should be costly both to submit and to retain, while supported and target-relevant claims should be worth preserving.

Our contributions are threefold.

- We expose claim-level contamination as a RAG safety failure that can be hidden by final-answer accuracy.
- We formulate hallucination control under corrupted evidence as objective-aware claim auditing with an explicit safety–coverage designer objective.
- We introduce OR-Wolf, a social-deduction-inspired evidence-trial protocol that audits claim survival before final answer generation.

Method

Overview

We propose **OR-Wolf**, an objective-aware audited claims game for hallucination-resistant retrieval-augmented reasoning. We formulate it as a finite sequential audited-claim protocol. The repeated extension induces a game in which agents adapt their submission strategies according to realized pay-offs.

Problem Formulation

Audited Claim Game. For each instance, let q denote the target scientific statement, $\mathcal{D}$ the retrieved document collection, and $\mathcal{N} = \{1, \dots, N\}$ the set of agents. Let the final outcome space be

$$\mathcal{Y} = \{\text{SUPPORTS}, \text{REFUTES}, \text{NEI}\}, \quad (1)$$

where NEI denotes NOT ENOUGH INFO. The protocol proceeds through four stages: evidence assignment, claim submission, peer challenge, and claim settlement and finalization.

Evidence Assignment. Nature assigns agent i an evidence-integrity type $\theta_i \in \Theta$ and constructs an agent-specific evidence bundle

$$D_i = \Gamma(\mathcal{D}, q, \theta_i), \quad \sum_{i=1}^N \mathbb{I}[\theta_i = \text{CORRUPTED}] = K, \quad (2)$$

where Γ is the evidence-assignment procedure and K controls evidence corruption.

Agent i initially observes only q and its own bundle D_i . At later stages, it also observes the public audit trace accumulated so far, but never the complete integrity profile $\theta = (\theta_1, \dots, \theta_N)$.

Claim Submission. Each agent independently submits an answer, answer confidence, and a set of atomic claims:

$$\begin{aligned} a_i^{\text{sub}} &= (y_i, \rho_i, C_i), \\ C_i &= \{c_{i1}, \dots, c_{im_i}\}, \\ c_{ij} &= (\text{id}_{ij}, x_{ij}, e_{ij}, p_{ij}, i). \end{aligned} \quad (3)$$

Here, $y_i \in \mathcal{Y}$ is the proposed answer, $\rho_i \in [0, 1]$ is the answer confidence, x_{ij} is the claim text, e_{ij} is its quoted evidence, $p_{ij} \in [0, 1]$ is the claim confidence, and i identifies the submitting agent.

In the current implementation, a claim inherits the answer direction of its submitting agent:

$$\ell(c_{ij}) = y_i, \quad y_i \in \{\text{SUPPORTS}, \text{REFUTES}\}. \quad (4)$$

Claims submitted under an agent-level NEI prediction do not contribute to either directional score during finalization.

All submitted claims form the public claim set

$$\mathcal{C} = \bigcup_{i=1}^N C_i. \quad (5)$$

OR-Wolf does not immediately aggregate the proposed answers $\{y_i\}_{i=1}^N$. Instead, it exposes their underlying atomic claims and evidence quotations for claim-level auditing.

Peer Challenge. After observing $\mathcal{C}$, agents may challenge claims submitted by other agents. A challenge issued by agent i against claim c_k is represented as

$$h_{ik} = (i, \text{id}(c_k), t_{ik}, r_{ik}, e_{ik}^h, \gamma_{ik}), \quad (6)$$

where t_{ik} is the challenge type, r_{ik} is the stated reason, e_{ik}^h is the challenger’s evidence, and $\gamma_{ik} \in [0, 1]$ is the challenge confidence.

Only challenges that target an existing claim and satisfy $\gamma_{ik} \geq \tau_{\text{ch}}$ enter the audit trace $\mathcal{H}_{\text{ch}}$. A challenge records a structured objection, but it does not directly delete a claim or determine its settled status.

The preceding stages produce an audited interaction trace, but do not determine which claims should survive or whether the system should answer or abstain. OR-Wolf therefore separates the interaction protocol from the designer’s retention and finalization rules.

Designer Objective. OR-Wolf uses a coverage-aware designer objective that balances final-answer reliability, claim-level safety, useful-evidence retention, and auditing cost. For compactness, let T, A, and B denote TARGETRELEVANTSUPPORTED, AMBIGUOUSTARGETSUPPORTED, and BACKGROUND-SUPPORTED, respectively, and let F, C, and U denote FABRICATEDQUOTE, CONTRADICTED, and UNSUPPORTED. We define

$$\begin{aligned} \mathcal{K}_{\text{bad}} &= \{U, C, F\}, \\ \mathcal{K}_{\text{useful}} &= \{T, A\}, \\ \mathcal{K}_{\text{sup}} &= \{T, A, B\}. \end{aligned} \quad (7)$$

For gold label y^* , final prediction $\hat{y}$, and claim-retention decisions $\{d_c\}_{c \in \mathcal{C}}$, the designer loss is

$$\begin{aligned} \mathcal{L} &= C_{\text{wrong}} \mathbb{I}[\hat{y} \neq y^*, \hat{y} \neq \text{NEI}] \\ &+ C_{\text{abs}} \mathbb{I}[\hat{y} = \text{NEI}, y^* \neq \text{NEI}] \\ &+ \sum_{c \in \mathcal{C}} d_c C_{\text{unsafe}}(\kappa_c) \mathbb{I}[\kappa_c \in \mathcal{K}_{\text{bad}}] \\ &+ C_{\text{del}} \sum_{c \in \mathcal{C}} (1 - d_c) \mathbb{I}[\kappa_c \in \mathcal{K}_{\text{useful}}] \\ &+ C_{\text{audit}} |H^{\text{ch}}|. \end{aligned} \quad (8)$$

The five terms penalize wrong non-abstaining predictions, false abstention, retained unsafe claims, deletion of decision-relevant supported claims, and auditing cost. Background-supported claims belong to $\mathcal{K}_{\text{sup}}$ but not $\mathcal{K}_{\text{useful}}$, because they are evidence-supported without directly resolving the target statement.

Because y^* is unavailable at inference time, Equation (8) is used for calibration and evaluation rather than as a test-time oracle. Test-time retention and finalization instead use the verifier-conditioned surrogate defined below.

Claim Settlement and Finalization. After peer challenge, each claim $c \in \mathcal{C}$ is evaluated against thereference evidence D^{ref} and target statement q :

$$\begin{aligned} (v_c, s_c) &= V_{\text{ev}}(x_c, e_c, D^{\text{ref}}), \\ (r_c, g_c) &= V_{\text{rel}}(x_c, q), \\ m_c &= M(x_c, e_c, q). \end{aligned} \quad (9)$$

Here, (v_c, s_c) is the evidence-verification label and confidence, (r_c, g_c) is the claim–target relation and confidence, and m_c is the maximum lexical overlap of the claim text or quoted evidence with the target statement. The evidence verifier evaluates claims against the clean reference evidence D^{ref} , rather than the potentially corrupted private bundle observed by the submitting agent.

Let

$$\begin{aligned} \mathcal{R}_{\text{tar}} &= \{\text{SUPPORTSTARGET}, \text{REFUTESTARGET}\}, \\ \mathcal{Y}_{\pm} &= \{\text{SUPPORTS}, \text{REFUTES}\}. \end{aligned} \quad (10)$$

The lexical-relevance gate is

$$z_c = \mathbb{I}[m_c \geq \tau_m \vee (\ell_c \in \mathcal{Y}_{\pm} \wedge m_c \geq \tau_w)], \quad (11)$$

where $\tau_m > \tau_w$ are the standard and weak lexical-overlap thresholds. The settled claim status is assigned by the ordered rule

$$\kappa_c = \begin{cases} v_c, & v_c \in \mathcal{K}_{\text{bad}}, \\ U, & v_c \neq \text{SUPPORTED} \vee s_c < \tau_{\text{ev}}, \\ T, & r_c \in \mathcal{R}_{\text{tar}} \wedge g_c \geq \tau_{\text{rel}}, \\ A, & z_c = 1, \\ B, & \text{otherwise.} \end{cases} \quad (12)$$

The cases are evaluated sequentially. Evidence failure takes priority; relation confidence then identifies explicit target relevance, and lexical overlap distinguishes ambiguous-target from background-supported claims.

The ledger stores each claim, its verifier signals, settled status, retention decision, and associated challenges. In the current implementation, claims are independently verified; peer challenges are retained for payoff accounting and audit analysis but are not inputs to Equation (12).

The filtering variants differ only in which supported statuses enter final regeneration. Evidence-Only retains $\mathcal{K}_{\text{sup}}$, Hard-Relevance retains only T, and Soft-Relevance retains $\mathcal{K}_{\text{useful}}$. All variants remove claims in $\mathcal{K}_{\text{bad}}$.

For objective-aware retention, define

$$\bar{p}_c = \max\{0.1, p_c\}, \quad \xi_c = 1 - \min\left\{1, \frac{s_c + m_c}{2}\right\}, \quad (13)$$

where $\bar{p}_c$ is the confidence used by the implementation and ξ_c measures the uncertainty of an ambiguous-target claim. The implemented local retention utility is

$$U_c = \begin{cases} \alpha_{\text{tar}} \bar{p}_c s_c, & \kappa_c = T, \\ \alpha_{\text{amb}} \bar{p}_c s_c - \omega_{\text{amb}} C_{\text{amb}} \xi_c, & \kappa_c = A, \\ \alpha_{\text{bg}} \bar{p}_c s_c - R_{\text{bg}}, & \kappa_c = B, \\ -C_{\text{fab}}, & \kappa_c = F, \\ -C_{\text{contra}} \bar{p}_c, & \kappa_c = C, \\ -C_{\text{unsup}} \bar{p}_c, & \kappa_c = U. \end{cases} \quad (14)$$

Supported claims receive status-dependent benefit, whereas verifier-defined bad claims receive negative utility. Here, C_{amb} is the uncertainty-risk scale used for ambiguous-target claims.

The retention decision is

$$d_c = \begin{cases} 0, & \kappa_c \in \mathcal{K}_{\text{bad}}, \\ 0, & \kappa_c = \text{B} \wedge b_{\text{bg}} = 0, \\ \mathbb{I}[U_c > \tau_{\text{keep}}], & \text{otherwise.} \end{cases} \quad (15)$$

Thus, verifier-defined bad claims are always excluded, while other eligible claims are retained when their utility exceeds

τ_{keep} .

Let

$$C^R = \{c \in C : d_c = 1\}.$$

Retained utility is grouped by the answer direction inherited from the submitting agent:

$$S_y = \sum_{\substack{c \in C^R \\ \ell_c = y}} \max\{0, U_c\}, \quad y \in \mathcal{Y}_{\pm}. \quad (16)$$

Let

$$S_+ = S_{\text{SUPPORTS}}, \quad S_- = S_{\text{REFUTES}}.$$

The provisional label and score summary are

$$\begin{aligned} \tilde{y} &\in \arg \max_{y \in \mathcal{Y}_{\pm}} S_y, \\ S_{\max} &= \max\{S_+, S_-\}, \\ \Delta &= |S_+ - S_-|. \end{aligned} \quad (17)$$

The directional conflict statistic is

$$R_{\text{conf}} = \begin{cases} 1, & S_+ + S_- = 0, \\ \frac{\min\{S_+, S_-\}}{S_+ + S_-}, & \text{otherwise.} \end{cases} \quad (18)$$

The answer utility is

$$U_{\text{ans}} = S_{\max} - C_{\text{wrong}} R_{\text{conf}}. \quad (19)$$

Here, R_{conf} measures insufficient or conflicting directional evidence.

The final prediction is

$$\hat{y} = \begin{cases} \text{NEI}, & U_{\text{ans}} \leq \tau_{\text{ans}}, \\ \text{NEI}, & \Delta < \tau_{\text{margin}}, \\ \tilde{y}, & \text{otherwise.} \end{cases} \quad (20)$$

The verifier therefore settles individual claims but never directly predicts the final label. Final prediction depends only on the directional utility of retained claims.

The matched designer-objective ablation separates three changes. Obj-Final applies Equation (20) to the claims retained by Soft-Relevance. Obj-Retain additionally applies Equation (15). Obj-Rescue is triggered only when objective-aware retention initially produces NEI and the count-based deletion-risk score exceeds its calibrated threshold. It restores deleted target-relevant and ambiguous-target supported claims and sufficiently target-overlapping background-supported claims before rerunning finalization.

Experimental Setup

We evaluate OR-Wolf on SciFact (Wadden et al. 2020) using three Qwen2.5-7B-Instruct agents and a DeBERTa-v3 verifier. We consider one- and two-agent corruption settings and compare against standard multi-agent debate, MUG-style agent filtering, and claim-level OR-Wolf variants. The primary metrics are accuracy, wrong non-abstention, false abstention, claim survival, supported-claim retention, and designer-objective loss. Objective weights are calibrated on 50 examples and evaluated on a disjoint 200-example set. Full prompts, hyperparameters, corruption construction, and evaluation procedures are provided in supplementary document.

Results and Analysis

Final-Answer Accuracy Masks Claim-Level Risk

Final-answer accuracy alone did not imply a factually reliable claim ledger. In the 50-example diagnostic pilot, single-agent RAG and self-consistency achieved answer accuracies of 96% and 94%, respectively. Yet neither method filtered verifier-flagged unsafe claims, yielding a 100% wolf-claim survival rate for both systems. Their retained-bad-claim losses were 0.88 and 2.62, respectively. The same failure mode appeared in standard multi-agent debate (MAD). By contrast, OR-Wolf’s verifier-based retention reduced verifier-defined wolf-claim survival to 0%. This comparison concerns claim filtering rather than end-to-end accuracy, because the pilot evidence conditions were not identical across methods.

Together, these results show that high final-answer accuracy can coexist with verifier-flagged unsafe claims in the intermediate evidence trace; because the pilot baselines received different evidence conditions, this comparison is diagnostic rather than a controlled method ranking. Moreover, claim status was assigned by the same verifier used by OR-Wolf., wolf-claim survival therefore should be interpreted as a verifier-conditioned diagnostic rather than an independently annotated hallucination rate.

Claim-Level Adjudication under Corrupted Evidence

Table 1 reports the main corruption-stress comparison on 200 SciFact examples with three agents. Wrong non-abstain (WNA) denotes an incorrect prediction made without abstention; false abstain (FA) denotes a NOTENOUGHINFO prediction when the gold label is SUPPORTS or REFUTES. WNA measures unsafe commitment, whereas FA captures excessive abstention.

Claim-level auditing reduced wrong non-abstaining predictions relative to answer-level voting, but much of this reduction came from abstention. With one corrupted agent, OR-Wolf-Soft increased accuracy by 6.0 percentage points, from 57.5% to 63.5%, and reduced WNA by 33.0 points, from 42.5% to 9.5%—a 77.6% relative reduction—compared with Standard MAD. Meanwhile, FA increased from 0.0% to 27.0%. With two corrupted agents, accuracy increased by 13.0 points, from 45.5% to 58.5%, and WNA decreased by 44.0 points, from 54.5% to 10.5%—an 80.7%

relative reduction—while FA increased from 0.0% to 31.0%. The WNA reductions therefore decomposed into higher accuracy and greater abstention: 6.0 and 27.0 points, respectively, with one corrupted agent, and 13.0 and 31.0 points with two. Net score nevertheless increased from 0.150 to 0.275 and from -0.090 to 0.210. Thus, OR-Wolf-Soft improved the safety–coverage trade-off primarily by replacing wrong non-abstaining predictions with abstentions, rather than by uniformly recovering the correct label. The comparison with agent-level filtering was considerably closer. OR-Wolf-Soft exceeded the MUG-style baseline by 1.0 accuracy point with one corrupted agent and by 0.5 points with two. The corresponding net-score gains were 0.020 and 0.015. MUG-style achieved slightly lower WNA rates, by 0.5 and 1.0 points, whereas OR-Wolf-Soft reduced FA by 1.5 points in both conditions. These results show that claim-level adjudication is competitive with agent-level filtering, but they do not establish a decisive advantage for either approach. We therefore characterize OR-Wolf as substantially better than answer-level MAD and comparable to the MUG-style baseline under the present evaluation.

The Safety–Coverage Trade-off

As shown above, the reduction in WNA was accompanied by substantial abstention. OR-Wolf-Soft abstained on 53.5% and 58.0% of examples, of which 27.0% and 31.0% were false abstentions. The gain therefore reflects selective risk control rather than uniform error correction.

The hard-versus-soft relevance ablation shows that aggressively claim filtering can improve caution at the cost of excessive evidence loss. In the 50-example pilot, hard relevance filtering achieved only 44% accuracy and abstained on 80% of examples. Soft relevance increased accuracy to 72% and reduced abstention to 40%. However, evidence-only filtering achieved 76% accuracy in the same pilot. The 200-example experiment similarly showed only small differences between evidence-only and soft-relevance filtering: 0.5 accuracy point with one corrupted agent and no accuracy difference with two. Soft relevance therefore reduces the excessive claim deletion and abstention caused by a hard relevance

gate. However, the results do not identify soft relevance as the primary source of OR-Wolf’s overall improvement. The stronger conclusion is that controlling claim survival is useful only when the retention rule preserves sufficient evidence for the final decision.

Matched Ablation of the Designer Objective

We next tested whether the designer objective improves the safety–coverage trade-off after claim auditing. We used the first 50 examples as a calibration set for the objective weights and decision thresholds. We then evaluated the subsequent 200 held-out examples with one corrupted agent. For each example, all four variants reused identical agent claims, peer challenges, and verifier settlements. This matched design isolates the effect of retention and finalization decisions from variation in sampled model outputs.

Compared with soft-relevance regeneration, objective-aware finalization increased observed accuracy and reduced observed WNA and designer-objective loss. Accuracy increased by 1.0 percentage point, from 62.0% to 63.0%, corresponding to two additional correct predictions among 200 examples. WNA decreased by 1.5 points, from 9.5% to 8.0%, corresponding to three fewer wrong non-abstaining predictions and a 15.8% relative reduction. Designer-objective loss decreased by 0.026, from 0.4762 to 0.4502, a 5.5% relative reduction. These changes also introduced one additional false abstention, increasing FA from 28.5% to 29.0%. The nature-style net score remained unchanged at 0.195; thus, the lower designer-objective loss did not translate into a higher score under the alternative utility measure. These values are point estimates, with paired statistical inference reported below.

Paired inference did not find the answer-level differences to be statistically distinguishable from zero. For accuracy, Soft alone was correct on seven examples, whereas Objective-Final alone was correct on nine; McNemar’s test yielded $p = 0.804$. For WNA, eight unsafe non-abstaining errors were removed, and five new ones were introduced ($p = .581$). For FA, two false abstentions were removed and three were introduced ($p = 1.000$). All paired 95% confidence intervals included zero. The observed effects are therefore directionally favorable but statistically uncertain at $n = 200$.

The loss decomposition shows that the favorable point estimate was driven by lower wrong-non-abstain cost. The average wrong-non-abstain component decreased from 0.190 to 0.160, while the false-abstain component increased from 0.228 to 0.232. Audit loss remained fixed at 0.0582 because the variants shared the same peer challenges and verifier settlements. Thus, the observed -0.026 loss difference consisted of a 0.030 reduction in wrong-non-abstain cost, partially offset by a 0.004 increase in false-abstention cost. Its paired-bootstrap 95% CI, $[-0.098, 0.042]$, nevertheless included zero. This pattern is consistent with the intended cost ordering of the designer objective.

Objective-aware retention did not further improve the answer metrics. It reduced supported-claim retention by 0.79 points relative to Soft, from 97.64% to 96.85%, and introduced an average good-deletion loss of 0.008. Rescue increased supported retention to 98.82%, 1.18 points above Soft and 1.97 points above objective retention. In count

Corrupt agents	Method	Acc. $\uparrow$	Net $\uparrow$	WNA $\downarrow$	FA $\downarrow$
1	Standard MAD	57.5%	0.150	42.5%	0.0%
	MUG-style filtering	62.5%	0.255	9.0%	28.5%
	OR-Wolf-Evid	63.0%	0.270	10.0%	27.0%
	OR-Wolf-Soft	63.5%	0.275	9.5%	27.0%
2	Standard MAD	45.5%	-0.090	54.5%	0.0%
	MUG-style filtering	58.0%	0.195	9.5%	32.5%
	OR-Wolf-Evid	58.5%	0.210	10.0%	31.5%
	OR-Wolf-Soft	58.5%	0.210	10.5%	31.0%

Table 1: Corruption-stress results on 200 SciFact examples with three agents. OR-Wolf-Evid retains verifier-supported claims, whereas OR-Wolf-Soft additionally applies soft target relevance. MUG-style filtering denotes an agent-level baseline.

Panel A. Matched Designer Objective Ablation

Variant	Acc. $\uparrow$	Obj. loss $\downarrow$	WNA $\downarrow$	FA $\downarrow$	Supp. ret. $\uparrow$
Soft	62.0%	0.4762	9.5%	28.5%	97.64%
Obj-Final	63.0%	0.4502	8.0%	29.0%	97.64%
Obj-Retain	63.0%	0.4583	8.0%	29.0%	96.85%
Obj-Rescue	63.0%	0.4502	8.0%	29.0%	98.82%

Panel B. Paired Inference: Obj-Final – Soft

Outcome	Difference	Paired-bootstrap 95% CI	Exact McNemar p
Accuracy	+1.0 pp	$[-3.0, +5.0]$ pp	.804
Wrong non-abstain	−1.5 pp	$[-5.0, +2.0]$ pp	.581
False abstain	+0.5 pp	$[-1.5, +2.5]$ pp	1.000
Designer-objective loss	−0.026	$[-0.098, +0.042]$	—

Table 2: Matched Designer Objective ablation and paired inference on 200 held-out SciFact examples with one corrupted agent. All variants reuse identical claims, peer challenges, and verifier settlements. Panel B reports paired differences defined as Obj-Final minus Soft. Confidence intervals use 20,000 paired percentile-bootstrap resamples; McNemar tests are exact and two-sided. Net score (0.195) and verifier-defined wolf-claim survival (0.0%) are omitted because they are constant across all variants.

terms, rescue halved the number of deleted verifier-supported claims relative to Soft, from six to three, and recovered five of the eight supported claims deleted by objective retention. The exploratory paired-bootstrap confidence intervals all included zero at the boundary: $[-2.56, 0.00]$ points for Retain versus Soft, $[0.00, 3.80]$ for Rescue versus Soft, and $[0.00, 5.04]$ for Rescue versus Retain. Because Rescue and retention produced identical answer-level predictions, the rescue step should be interpreted as a descriptive coverage-recovery safeguard rather than an answer-level improvement.

All four variants had zero verifier-defined wolf-claim survival because they shared the same verifier settlement and hard safety filter. The common zero wolf-claim survival rate therefore does not identify an additional contribution from the designer objective. With the audited ledger held fixed, objective-aware finalization reduced the observed designer loss from 0.4762 to 0.4502. However, the paired-bootstrap 95% CI included zero, so the data do not establish a reliable improvement. It also remains unknown whether this result holds under other objective weights.

Incentive Dynamics in the Repeated Claim Game

The repeated-game experiment examines whether claim-level payoffs encourage agents to change their strategy selection in the intended direction. Across 200 examples, the aggregate probability mass assigned to truthful strategies increased by 4.56 percentage points, from 40.46% to 45.01%, while the mass assigned to the aggressive strategy decreased by 3.12 points, from 19.89% to 16.77%. Mass assigned to the conservative abstention strategy also decreased by 2.81 percentage points, from 19.77% to 16.96%. The simultaneous increase in truthful strategy mass and decrease in both aggressive and abstention strategies suggests that the observed shift was not driven solely by more evasive behavior. The balanced claimant showed the largest individual increase, rising by 5.32 points from 20.57% to 25.89%. (Figure 1).

These shifts are consistent with the intended incentive de-

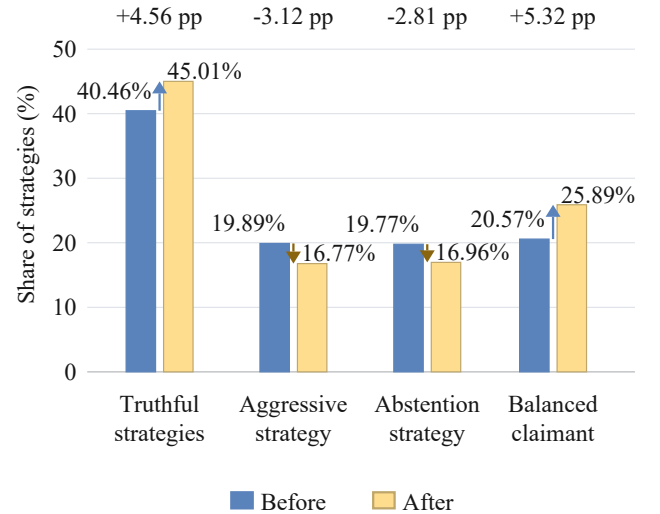


Figure 1: Strategy-mass changes during the 200-example repeated audited claim game. Balanced claimant is also shown individually; grouped and individual rows should not be summed.

sign: strategies that submit useful evidence-grounded claims became more attractive, while aggressive claim submission became less attractive. However, it does not establish that agents learned to be truthful or that repeated learning improved final task performance. OR-Wolf-Repeat achieved 64.5% accuracy and a 0.280 net score, but a matched 200-example no-learning control was not completed. Calibration error also remained high at 0.5287, and the proper-scoring and fabrication-penalty ablations were incomplete. We therefore interpret the strategy trajectory as preliminary mechanism evidence that payoff feedback can alter strategy selection, not as causal evidence of improved factuality.

Claim Auditing and Failure Analysis

Under verifier-defined status, OR-Wolf-Soft achieved 0% wolf-claim survival while retaining 97.05% and 97.26% of supported claims with one and two corrupted agents. Challenge precision was 77.86% and 72.73%, but challenge recall fell from 38.80% to 21.76% as corruption increased. Thus, peer challenges were generally well targeted when they occurred, but they covered only a minority of claims later classified as bad by the verifier.

The low challenge recall reveals an important mechanism boundary. The current results suggest that much of the observed filtering performance may come from verifier settlement and the retention rule rather than from peer cross-examination. Challenges may still provide useful local evidence and agent-facing incentives, but the present experiments do not isolate their incremental contribution to final-task performance. A no-challenge ablation is needed before attributing the reliability gain to peer auditing itself.

The verifier sanity check serves as a measurement-validity test because the same verifier both provides supplies the settlement signal used by OR-Wolf and defines the reported claim-safety diagnostics. It achieved 66% accuracy on 100 examples, with approximately 46.2% recall on REFUTES cases. This limited recall suggests that the verifier may under-detect some contradicted evidence. Consequently, zero verifier-defined wolf-claim survival cannot be interpreted as zero ground-truth hallucination. Independent human claim annotation or evaluation with a held-out verifier is required to determine how often the retained ledger is genuinely factual.

Summary of Findings

The experiments support three conclusions at different levels of strength. First, claim-level adjudication substantially reduced wrong non-abstaining answers relative to answer-level MAD under corrupted evidence. With relative reductions of 77.6% and 80.7 under one and two corrupted agents, respectively. Second, its advantage over the MUG-style agent-filtering baseline was small, supporting comparability rather than decisive superiority. Third, in the matched held-out ablation, objective-aware finalization reduced WNA by 15.8% relative and designer-objective loss by 5.5% in point estimates. Although the point estimates favored objective-aware finalization, all paired confidence intervals included zero and the exact McNemar tests were not significant. The effect therefore remains statistically uncertain at $n = 200$. Objective-aware retention did not independently improve answer-level gain, while rescue descriptively mitigated its evidence-deletion risk. Finally, strategy movement in the repeated game and observed challenge behavior are mechanism-level signals whose independent contribution to factual accuracy remains to be established.

Analysis

Claim safety is distinct from answer accuracy. The diagnostic results show that a correct final label can coexist with verifier-flagged claims in the retained evidence trace.

Under corrupted evidence, OR-Wolf reduced wrong non-abstaining predictions relative to standard MAD, but much of this reduction came from replacing unsafe commitments with abstentions. Final-answer accuracy and retained-claim reliability therefore capture different aspects of factuality and should be evaluated together.

Settlement and finalization drive the main performance gains. The small differences between OR-Wolf and MUG-style filtering do not establish that claim-level adjudication is uniformly superior to agent-level filtering. Moreover, limited challenge recall and the shared zero wolf-claim survival across objective variants suggest that verifier settlement and the hard safety filter account for much of the observed claim removal. The matched ablation further indicates that objective-aware finalization selects a more risk-sensitive operating point, although the observed improvements remain statistically uncertain. Rescue is thus better viewed as a coverage safeguard than as an independent source of answer-level gains.

The method controls risk when trustworthy evidence remains available. Because both settlement and evaluation depend on the verifier, zero verifier-defined wolf-claim survival does not imply zero ground-truth hallucination. Claim filtering also cannot recover evidence that is missing from every agent’s input. When truthful coverage collapses, increased abstention is the appropriate response. The strongest conclusion supported by the current experiments is therefore that OR-Wolf provides explicit control over which claims survive and when the system should abstain, but it does not guarantee factual recovery under arbitrary corruption.

Limitations

The current evaluation is limited to SciFact and synthetic counterfactual corruption, although the main comparisons use 200 examples. Claim-safety results are verifier-conditioned: the verifier achieved 0.66 accuracy and 46.2% recall on REFUTES cases, and the same verifier both settles claims and defines wolf-claim status. Because of computational constraints, all experiments use Qwen2.5-7B-Instruct with three agents; performance with larger models, more agents, and broader domains remains untested. Finally, OR-Wolf assumes access to truthful reference evidence and cannot guarantee coverage when all private evidence is corrupted, in which case abstention is expected.

Conclusion

This paper presents OR-Wolf, an objective-aware claim-auditing framework for hallucination control in RAG under corrupted evidence. By shifting the unit of adjudication from answers to individual claims, OR-Wolf enables more precise control over unsupported claims while preserving useful evidence. Our experimental results show that claim-level adjudication substantially reduces wrong non-abstaining errors compared with standard multi-agent debate and performs competitively with agent-level filtering. These results demonstrate the value of claim-level auditing for improving the reliability of multi-agent RAG.

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